

Celestial Simulation — Project Vision

Compiled from Greg's input across multiple conversations. This document captures the user's vision and requirements, not implementation decisions or Claude's suggested approaches.

Core Concept

A browser-based celestial simulation that serves three overlapping purposes: a night sky planetarium, a geometric data explorer for celestial relationships, and a personal astrology tool. It should run on a phone, be hostable on GitHub Pages, and prioritize simplicity, accuracy, and a minimal/cartoonish aesthetic over photorealism.

The View: Standing Inside the Celestial Sphere

The fundamental experience is standing on the surface of the Earth and looking up at the night sky — a 360° view of the interior of the celestial sphere.

- The perspective is approximately 5 feet above the Earth's surface, as if you're a person standing outside.
- The viewport is a portrait-oriented phone screen — a rectangular window into the spherical sky.
- The user can look in any direction by dragging — up at the zenith, around the horizon, and even down at their feet. The full celestial sphere should be traversable.
- The default orientation is centered on the zenith (straight up).
- 90% of the time, users will be looking at the upper hemisphere (above the horizon), but they should be able to follow a planet's path below the horizon and "under their feet."
- The entire celestial sphere interior surface doesn't need to be visible at once on the phone canvas — it should be fully traversable by dragging/panning to center any point.
- Think of it as a non-Euclidean 2D star map projected onto the interior surface of a 3D sphere, viewed from inside.

The Horizon and Below

- The surface of the Earth acts as a semi-transparent occlusion — roughly 80% alpha-reduced, so stars and planets below the horizon are dimmed but still visible.
- The Earth could be rendered as a sphere (scaled in a cartoony way, like a small asteroid à la The Little Prince) or treated as a flat plane tangent to the surface — this is an open design question.
- No rendering of the user's body, feet, or ground details — just the semi-transparent earth/ground plane as a visual divider between above-horizon and below-horizon sky.

Earlier Exploration: Viewing From Center vs. Surface

An earlier consideration: the user could be located at the center of the celestial sphere (simpler) or on the external surface of a smaller sphere (Earth) inside the larger celestial sphere. The parallax between these is effectively nothing for stars and negligible for planets at this resolution, so treating the observer as being at the center of the celestial sphere is fine for now. But the surface-of-Earth framing matters for the horizon — what's visible depends on where you're standing.

What's Rendered

Stars (Milestone 1)

- The ~1,000 brightest stars, accurately positioned on the celestial sphere.
- Low resolution is fine — this is meant to run on a phone without needing to zoom in.
- Stars are essentially static over the relevant time range (proper motion is negligible), but the coordinate frame itself drifts due to precession.

Sun and Moon (Milestone 2)

- Sun position is important for defining day/night and for astrological context.
- Moon position also needed eventually — acknowledged as the hardest single body to compute accurately.
- Moon can be deferred to a later milestone if needed.

Planets (Milestone 2–3)

- The 7 naked-eye planets (Mercury, Venus, Mars, Jupiter, Saturn, plus Uranus and

Neptune if desired).

- Positions should be accurate enough to be visually correct on a phone screen — arcsecond precision is overkill, arcminute-level is more than sufficient.

Overlays (Later Milestones)

- **Zodiac divisions:** The sky divided into the 12 traditional zodiac sections — these are angular wedges along the ecliptic.
 - **88 classical constellations:** Optional overlay showing IAU constellation boundaries and/or traditional asterism lines.
 - **Ecliptic line, celestial equator, horizon compass labels (N/S/E/W):** Reference lines for orientation.
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Interaction and Controls

Observer Position

- The simulation places you at a specific latitude/longitude on Earth.
- Evanston, IL (41.88°N, 87.68°W) as the default/testing location.
- Eventually: user-settable location.

Time

- Should be able to show the sky for any minute/hour/day/year within the supported time range.
- Time range: 1900 AD to 2100 AD for the initial version. (Originally considered 2000 BC to 2500 AD but scoped down for accuracy and simplicity.)
- Time controls: date/time picker, and a “play” button to advance time and watch the sky move.

Viewport Navigation

- Touch-drag to look around the celestial sphere from the observer’s position.
 - The phone screen is a window into the sky — you pan it around rather than seeing the whole sky at once.
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The Personal / Astrology Layer

This is the most speculative and creatively open part of the project.

Birth Chart as a Point in the Sky

- Capture a birth datetime and location.
- Compute and cache the state of the sky at that moment — positions of all planets, sun, moon.
- This birth sky becomes a personal reference frame.

Comparative Visualization Ideas

- Show the relationship between the birth sky and the current sky.
- Example concept: if the night sky the moment you were born had Jupiter maximally far from your zenith, you'd almost always be "below your birth Jupiter level" — and the rare moments when Jupiter is closer to your current zenith than it was at birth would be notable.
- Could show distance of each planet from your birth-sky position vs. current position.
- Could represent the user's birth identity as a single point in the night sky and show how current planetary positions relate to it.
- The goal is NOT to reproduce traditional astrological interpretations — it's to create a tool where you can see the raw geometric relationships and create your own meaning.
- "I don't actually know what Venus rising means, and maybe I don't need to. If I could see the relative positions of all the planets on the night sky interior globe, I could make up my own meaning and headcanon and emotive intuitions about how the current situation is different or particular for my birth sign."
- Leave interpretation to the user — the tool shows the geometry, the user creates the meaning.

Open Questions (User's Own)

- What's the most intuitive geometric relationship to visualize between birth sky and current sky?
 - Is angular distance from zenith the right metric, or is there something more intuitive?
 - Could there be multiple relationship views to toggle between?
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Technical Preferences and Constraints

Platform

- Browser-based web app, single HTML file with vanilla JS (no build step).
- Hostable on GitHub Pages.
- Must run smoothly on a modern phone (primary target device).
- Portrait orientation assumed as primary.

Rendering

- Low resolution is fine — meant to be viewed on a phone screen without zooming.
- Minimal, simplistic, almost cartoony aesthetic — not photorealistic.
- Stars as dots, planets as slightly larger/colored dots, sun and moon with distinct markers.

Data Budget

- Stars: ~1,000 entries with RA, Dec, magnitude, optionally name/constellation. Static data, maybe 50–80 KB as JSON.
- Planets: dynamic computation preferred over pre-computed ephemeris tables — dramatically less storage (~50–100 KB of coefficients vs. tens of MB of tables).
- Overall: the simulation should be lightweight enough to load and run on a phone without significant data transfer.

Astronomy Engine

- Start with an existing JS astronomy library (e.g., astronomy-engine) for planet/sun/moon positions and coordinate transforms.
- Possibly replace with custom implementation later for more control.
- The library doesn't include a star catalog — that's separate data.

Prior Art and Experience

- Greg has extensive experience with PIXI.js and Three.js for browser-based simulations.
- Has a working Three.js prototype already built (referenced in earlier conversations).
- Has explored sphere-to-2D-grid mapping extensively in past conversations (variable-

width row grids, Platonic solid mappings, cubemaps).

- Other simulation projects: Simplant (cellular automata plant evolution), rgbfields (RGB ecosystem simulation), absorption (particle-based plant growth), lava lamp simulations (LBM, metaballs, raymarching).

Aesthetic and Philosophical Notes

- The tool should feel like it surfaces real geometric relationships that are always there but normally invisible.
- The astrology layer isn't about believing in astrology — it's about using the actual geometry of the solar system as a framework for personal meaning-making.
- The planetarium function grounds it in physical reality; the astrology layer opens it to personal interpretation.
- Half planetarium, half personal oracle — but the oracle shows you real data and lets you decide what it means.